

INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: Polynomial Regression

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LEAST-SQUARES FIT OF A STRAIGHT LINE

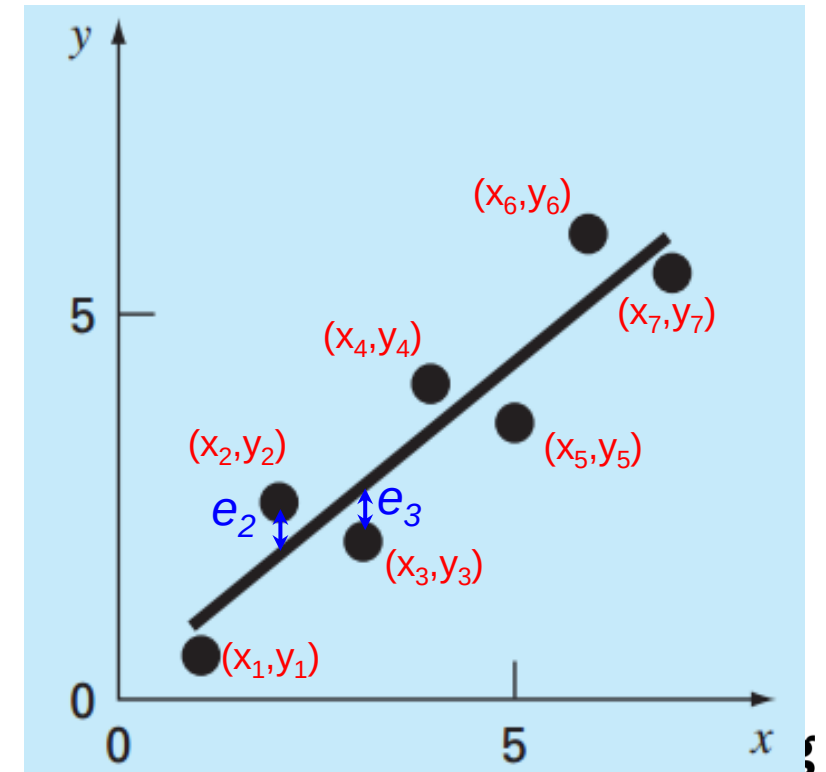
$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i) = 0$$

$$\frac{\partial S_r}{\partial a_1} = -2 \sum [(y_i - a_0 - a_1 x_i) x_i] = 0$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$



ERROR OF LINEAR REGRESSION

Sum of the Squared Error

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

Standard Error of the Estimate ($s_{y/x}$)

$$s_{y/x} = \sqrt{\frac{S_r}{n-2}}$$

← Degree of freedom (for data used in linear regression)

The coefficient of determination

$$r^2 = \frac{S_t - S_r}{S_t}$$

Improvement or error reduction due to the linear regression

Error before the linear regression

r : correlation coefficient

$$r = \sqrt{\frac{S_t - S_r}{S_t}}$$

POLYNOMIAL REGRESSION

Polynomial Regression

- Fitting a curve to a set of paired observations
 $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$

A Curve for Regression (2nd order Polynomial)

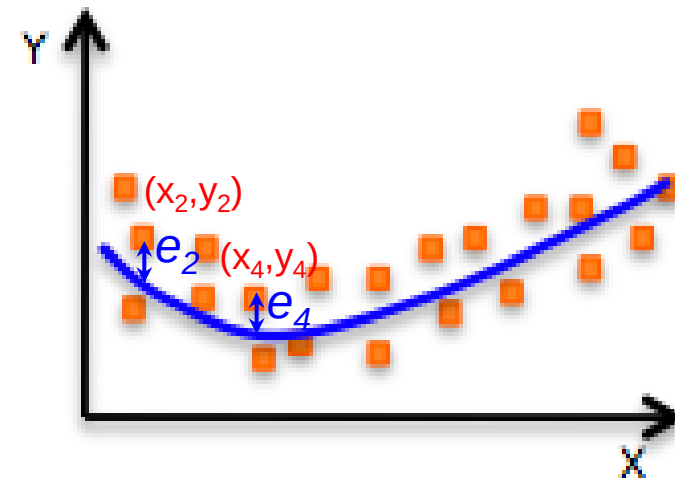
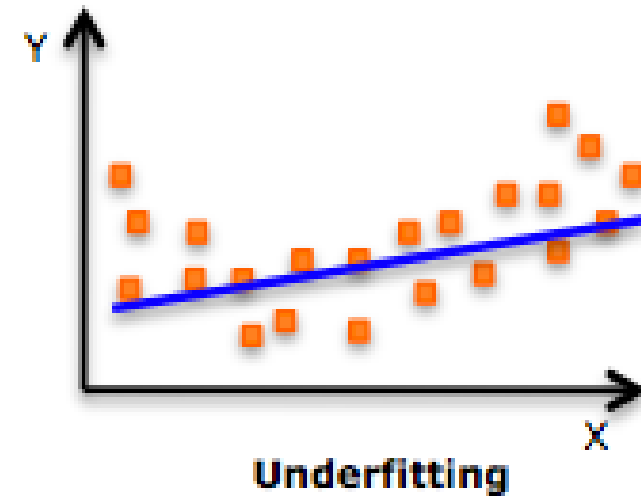
$$y = a_0 + a_1x + a_2x^2 + e$$

Error/residual

Residual

- Discrepancy between the true value of y and the predicted value predicted by the 2nd order polynomial

$$y_{\text{predicted}} = a_0 + a_1x + a_2x^2$$



CRITERION FOR FIT

Sum of the Squares of the Residual (2nd Order Polynomial)

- Measured y_i vs. predicted $y_{predicted} = a_0 + a_1x + a_2x^2$

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)^2$$

$$S_r = \sum_{i=1}^n (y_i - y_{predicted})^2$$

- Squares ensure no “cancelling” of error/residual while providing smooth derivative
- Yield a unique curve for a given set of data

Goal: Identify the best combination of a_0 , a_1 , and a_2 yielding minimum “sum of the squares of the residual”

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

Sum of the Squares of the Residual

- Measured y_i vs. predicted $y_{\text{predicted}} = a_0 + a_1x + a_2x^2$

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)^2$$

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1x_i - a_2x_i^2)$$

- Differentiate “each individual term” in “ Σ ” with respect to a_0
- Use the “chain rule”

$$\begin{aligned} \frac{\partial S_r}{\partial a_0} &= \sum_{i=1}^n \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)^2}{\partial a_0} = \sum_{i=1}^n 2(y_i - a_0 - a_1x_i - a_2x_i^2) \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)}{\partial a_0} \\ &= 2 \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)(-1) = -2 \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2) \end{aligned}$$

- Differentiate “each individual term” in “ Σ ” with respect to a_1
- Use the “chain rule”

$$\frac{\partial S_r}{\partial a_1} = -2 \sum x_i (y_i - a_0 - a_1x_i - a_2x_i^2)$$

$$\begin{aligned} \frac{\partial S_r}{\partial a_1} &= \sum_{i=1}^n \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)^2}{\partial a_1} = \sum_{i=1}^n 2(y_i - a_0 - a_1x_i - a_2x_i^2) \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)}{\partial a_1} \\ &= 2 \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)(-x_i) = -2 \sum_{i=1}^n x_i (y_i - a_0 - a_1x_i - a_2x_i^2) \end{aligned}$$

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

Sum of the Squares of the Residual

- Measured y_i vs. predicted $y_{\text{model}} = a_0 + a_1x + a_2x^2$

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)^2$$

$$\frac{\partial S_r}{\partial a_2} = -2 \sum x_i^2 (y_i - a_0 - a_1x_i - a_2x_i^2)$$



- Differentiate “each individual term” in “ Σ ” with respect to a_2
- Use the “chain rule”

$$\begin{aligned} \frac{\partial S_r}{\partial a_2} &= \sum_{i=1}^n \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)^2}{\partial a_2} = \sum_{i=1}^n 2(y_i - a_0 - a_1x_i - a_2x_i^2) \frac{\partial (y_i - a_0 - a_1x_i - a_2x_i^2)}{\partial a_2} \\ &= 2 \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2) (-x_i^2) = -2 \sum_{i=1}^n x_i^2 (y_i - a_0 - a_1x_i - a_2x_i^2) \end{aligned}$$

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

$$\sum_{i=1}^n y_i - \sum_{i=1}^n a_0 - \sum_{i=1}^n a_1 x_i - \sum_{i=1}^n a_2 x_i^2 = 0$$

Constant Constant

$$(n)a_0 + \left(\sum x_i\right)a_1 + \left(\sum x_i^2\right)a_2 = \sum y_i$$

$$\frac{\partial S_r}{\partial a_1} = -2 \sum x_i (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

$$\sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i a_0 - \sum_{i=1}^n a_1 x_i^2 - \sum_{i=1}^n a_2 x_i^3 = 0$$

$$\left(\sum x_i\right)a_0 + \left(\sum x_i^2\right)a_1 + \left(\sum x_i^3\right)a_2 = \sum x_i y_i$$

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

$$\frac{\partial S_r}{\partial a_2} = -2 \sum x_i^2 (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$



$$\sum_{i=1}^n x_i^2 y_i - \sum_{i=1}^n x_i^2 a_0 - \sum_{i=1}^n a_1 x_i^3 - \sum_{i=1}^n a_2 x_i^4 = 0$$



$$\left(\sum x_i^2 \right) a_0 + \left(\sum x_i^3 \right) a_1 + \left(\sum x_i^4 \right) a_2 = \sum x_i^2 y_i$$

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

Three Equations and Three unknowns: a_0 , a_1 , and a_2

$$\begin{cases} (n)a_0 + (\sum x_i)a_1 + (\sum x_i^2)a_2 = \sum y_i \\ (\sum x_i)a_0 + (\sum x_i^2)a_1 + (\sum x_i^3)a_2 = \sum x_i y_i \\ (\sum x_i^2)a_0 + (\sum x_i^3)a_1 + (\sum x_i^4)a_2 = \sum x_i^2 y_i \end{cases}$$

- The coefficients of the unknowns can be calculated directly from the observed data (x_i, y_i)
- A least-squares regression of the 2nd-order polynomial is equivalent to solving a system of three simultaneous linear equations
- Standard Error

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$

$$s_{y/x} = \sqrt{\frac{S_r}{n-3}}$$

For 2nd Order Polynomial Regression: we need

$$\left\{ \begin{array}{l} \sum_{i=1}^n x_i y_i \quad \sum_{i=1}^n x_i^2 y_i \\ \sum_{i=1}^n x_i^2 \quad \sum_{i=1}^n x_i^3 \quad \sum_{i=1}^n x_i^4 \\ \sum_{i=1}^n x_i \quad \text{or} \quad \bar{x} \\ \sum_{i=1}^n y_i \quad \text{or} \quad \bar{y} \end{array} \right.$$

AN EXAMPLE

2nd order polynomial least squares regression of the x and y values in the first 2 columns

x_i^4	x_i^3	x_i^2	x_i	y_i	$x_i y_i$
0	0	0	0	2.1	0
1	1	1	1	7.7	7.7000
16	8	4	2	13.6	27.2000
81	27	9	3	27.2	81.6000
256	64	16	4	40.9	163.6000
625	125	25	5	61.1	305.5000
979	225	55	Σ	152.6	

Solution:

First necessary quantities for regression will be computed:

$$\left\{ \begin{array}{l} n = 6 \quad \sum x_i y_i = 585.6 \quad \sum x_i^2 y_i = 2488.8 \\ \sum x_i = 15 \quad \bar{x} = \frac{15}{6} = 2.5 \quad \sum x_i^2 = 55 \quad \sum x_i^3 = 225 \quad \sum x_i^4 = 979 \\ \sum y_i = 152.6 \quad \bar{y} = \frac{152.6}{6} = 25.433 \end{array} \right.$$

$$(n)a_0 + (\sum x_i)a_1 + (\sum x_i^2)a_2 = \sum y_i$$

$$(\sum x_i)a_0 + (\sum x_i^2)a_1 + (\sum x_i^3)a_2 = \sum x_i y_i$$

$$(\sum x_i^2)a_0 + (\sum x_i^3)a_1 + (\sum x_i^4)a_2 = \sum x_i^2 y_i$$

$$6a_0 + 15a_1 + 55a_2 = 152.6$$

$$15a_0 + 55a_1 + 225a_2 = 585.6$$

$$55a_0 + 225a_1 + 979a_2 = 2488.8$$

AN EXAMPLE

2nd order polynomial least squares regression of the x and y values in the first 2 columns

x_i	y_i
0	2.1
1	7.7
2	13.6
3	27.2
4	40.9
5	61.1
Σ	152.6

Solution:

The linear equations are then given by:

$$\begin{aligned} (n)a_0 + (\sum x_i)a_1 + (\sum x_i^2)a_2 &= \sum y_i \\ (\sum x_i)a_0 + (\sum x_i^2)a_1 + (\sum x_i^3)a_2 &= \sum x_i y_i \\ (\sum x_i^2)a_0 + (\sum x_i^3)a_1 + (\sum x_i^4)a_2 &= \sum x_i^2 y_i \end{aligned} \Rightarrow \begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{Bmatrix} \Rightarrow \begin{cases} a_0 = 2.47857 \\ a_1 = 2.35929 \\ a_2 = 1.86071 \end{cases}$$

Gauss Elimination or LU
Decomposition

AN EXAMPLE

2nd order polynomial least squares regression of the x and y values in the first 2 columns

x_i	y_i	$(y_i - \bar{y})^2$	$y_{predict}$	$(y_i - a_0 - a_1x_i - a_2x_i^2)^2$
0	2.1	544.44	$\vdots$	0.14332
1	7.7	314.47	$\vdots$	1.00286
2	13.6	140.03	$\vdots$	1.08158
3	27.2	3.12	$\vdots$	0.80491
4	40.9	239.22	$\vdots$	0.61951
5	61.1	1272.11	$\vdots$	0.09439
Σ	152.6	2513.39		3.74657
		Sum of squared residual		Sum of squared error

Solution:

Error of the 2nd-order Polynomial Regression:

$$S_r = \sum_{i=1}^n (y_i - y_{predicted})^2$$

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)^2$$

The standard error of the estimate:

$$s_{y/x} = \sqrt{\frac{S_r}{n-3}} = \sqrt{\frac{3.74657}{6-3}} = 1.2$$

$$r^2 = \frac{S_t - S_r}{S_t}$$

$$r^2 = \frac{2513.39 - 3.74657}{2513.39} = 0.99851$$

LEAST-SQUARES REGRESSION OF AN MTH ORDER POLYNOMIAL

$$y = a_0 + a_1x + a_2x^2 + \cdots + a_mx^m + e$$

$$S_r = \sum_{i=1}^n \left(y_i - a_0 - a_1x - \cdots - a_mx^m \right)^2$$

- The approach is similar to that for the 2nd order polynomial
- Take derivative of S_r with respect to $a_0, a_1, a_2, \dots, a_m$
- Set all the derivatives to zero and then solve a system of $m+1$ linear equations

$$s_{y/x} = \sqrt{\frac{S_r}{n - (m + 1)}}$$

Need at least $m+1$ data points to uniquely determine the coefficients $a_0, a_1, \dots, a_m$. $m+1$ degrees of freedom is lost